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1. Introduction:

This project analyzes chess games. Each record represents one full game, including information about player ratings, game type (time control), opening used, and several variables describing the game's, such as piece advantages and capture counts.

The primary goal of this project is to predict probability of white winning a chess game by model that estimates whether the White player will win based on different parameters of the game.

1.2 Dataset Overview:

The dataset contains a mixture of numeric and categorical variables. Each game contain the game type, who won, what opening they used, and some numeric variables that describe the difference between white and Black scores and ratings and piece counts.

1.3 What Makes This Project Interesting?

Chess is a very strategic game with a lot of tactics, openings, and time management yet there are more aspects of the game are not directly measurable. This dataset gives us only some of this information of the gameplay:

- Rating differences
- Opening choices
- Material advantages (queen, rook, bishop, knight, pawn)
- Tactical behavior (checks and captures)

Despite that, can we still predict the winner?

To me Certain variables stand out immediately:

- Does difference in queen number (`queen_count_diff ≥ 1`) guarantees a win.
- Does rating difference significantly increase the expected win probability

This will provide insights into how much the observable, recorded aspects of the game influence the result and how much remains driven by unrecorded strategic skill.

What I am trying to do with this dataset is to predict who wins based on their rating and what opening they are using and their play style in general and how significant each factor is.

For example, losing a queen almost always indicates a losing position, however losing a bishop versus a knight may not. Lower-rated players often lose because of tactical blunders, while at higher levels openings and positional play matter more. At the end, this project tries to answer:

Given these variables, how accurately can we predict who will win the game?

2. Data Exploration and Preparation:

Dataset Columns:

- `game_type`: A categorical variable 4 levels indicate the speed/time control of the game. Levels: Classical, Blitz, Bullet, and Correspondence.
- `whiteWin`: A Boolean variable TRUE if White won; FALSE if Black won.
- opening: A categorical variable, 269 levels indicate the opening name, representing the sequence of initial moves used in the game.
- `elo_diff`: A numeric variable Rating shows the difference: white rating – black rating. A positive value means White has the higher rating.
- `pawn_count_diff`: A numeric variable shows the difference in remaining pawns. Positive means White has more pawns.
- `bishop_count_diff`: a numeric variable that shows the difference in number of bishops. Positive means White has more bishops.
- `knight_count_diff`: A numeric variable that shows the difference in the number of knights. Positive means White has more knights.
- `rook_count_diff`: A numeric variable that shows the difference in the number of rooks. Positive means White has more rooks.
- `queen_count_diff`: A numeric variable that shows the difference in the number of queens. Positive means White has more queens.
- `captured_score_diff` A numeric variable shows the difference in material score based on piece values (pawn = 1, bishop/knight = 3, rook = 5, queen = 9). Positive means White captured more valuable material.
- `capture_diff`: A numeric variable shows the difference in total captured pieces. Positive means White captured more pieces.
- `check_diff`: A numeric variable shows the difference in number of checks. Positive means White checked Black more often.

The dataset contains 117,148 rows (one per game) and 12 variables of chess game records collected from the Lichess online chess platform, a publicly available chess database.

Each row represents a completed chess game, with information on:

- Game metadata (game type, opening used)
- Player rating information (rating difference)

- Game outcome (White/Black win)
- Board state differences (piece material and activity)

2.1 Preprocessing and Data Cleaning

Several steps were performed to clean and prepare the data.

- Dropped rows have the Draw outcome for our model; now we can only work with variables that have 2 states
- Converting categorical variables to factors (`game_type`, `opening`)
- Dropped rows that have missing critical values (e.g., missing outcome, rating)
- Create the different columns to reflect the in-game advantages.

2.2 Summary Statistics and Visualizations

- Mean, Median shows that most of the player were evenly matched.
- Blitz was the most common game type.

3. Model Building

Based on our response variable defined as `whiteWin = 1` when white wins and `whiteWin = 0` otherwise given its binary nature a logistic regression model was chosen implemented using the `glm()` function with binomial family. Based on that, the logistic regression model is:

$$\text{logit}(\hat{p}) = \beta_0 + \beta_1 \text{gameType} + \beta_2 \text{opening} + \beta_3 \text{eloDiff} + \beta_4 \text{pawnCountDiff} + \beta_5 \text{knightCountDiff} + \beta_6 \text{bishopCountDiff} + \beta_7 \text{rookCountDiff} + \beta_8 \text{queenCountDiff} + \beta_9 \text{capturedScoreDiff} + \beta_{10} \text{capturedDiff} + \beta_{11} \text{checkDiff}$$

After building this model, we will look at the model summary to have some detailed statistics of our fitted model, allowing us to assess how well the predictors explain the outcome. We can see that we have an NA coefficient for one of the predictors, `captured_diff`. This means that this variable has a perfect collinearity with another predictor. Therefore, I proceeded to evaluate the model for potential multicollinearity, and examining the data to see if we have Influential observations.

3.1 Checking for Influential Observations

I Used the studentized residuals `rstudent(model)` and identified cases whose residuals exceeded the critical value determined by a very stric signifcance level $\alpha = 0.0001.$ therefore any observations above this threshold were considered potential outliers with this test game_ids(`106458, 71665, 71101, 51986, 46879, 39701, 27616, 21640, 6688`) but before removing these observation I checked the dataset directily to see if these value represented an error or miscoded data or simply rare but valid scenario. Eventhough some of them are very rare but are still valid observation, Therefore I won't be removing any of them.

3.2 Checking for Multicollinearity

Using VIF from regclass library on the model showed that we have serval variables that have high GVIF score `pawn_count_diff`, `queen_count_diff`, `bishop_count_diff`, `knight_count_diff`, `rook_count_diff`, and `captured_score_diff`. > 1 That indicate that we have an issue of Multicollinearity and it's the case because some of these variable are calculated using one another. We will fix it when we use elastic net to choose our best model it will reduce or remove the variable with the multicollinearity.

3.3 Variable selection Best Model

I used a K-Fold cross validation $k = 5$ to find the $\hat{\lambda}$, thet value of λ that minimize the $\widehat{MSPE}$ the most In Elastic Net. After running `cv.glmnet()` multiaple times with foldid of 5 and α value from 0 to 1 with 0.05 increment and with a binomail family I found the best α value that has the lowest λ and it was $\alpha = 1$ meaning pure LASSO provides the best predictive preformance for this dataset. So after apply this alpha again into elstic net function we get the variables for our best model.

$$\begin{aligned}\mathrm{logit}(\hat{p}) = \beta_0 &+ \beta_1\mathrm{gameType} + \beta_2\mathrm{opening} + \beta_3\mathrm{eloDiff} + \beta_4\mathrm{queenCountDiff}\\ &+ \beta_5\mathrm{bishopCountDiff} + \beta_6\mathrm{capturedScoreDiff} + \beta_7\mathrm{capturedDiff} + \beta_8\mathrm{checkDiff}\end{aligned}$$

Around half of the opening had zero coefficients and some of game type .Also It shows that some of variable that had Multicollinearity were removed while some remain. Specialy the capturedDiff that had perfict collinearity was kept in this new model.

3.4 Prediction intervals.

Let's try to make some prediction using the estimates of λ to calcuate the prediction line for the chess games via Elastic net regression. Using the predict function. We add our elastic net regresion model with a matrix for our dataset model this will give us a prediction for each game in the dataset.

To evaluate model performance on new data, a one of my games was constructed:

elo_diff: +8 (White slightly higher rated), pawn_count_diff: +1, queen_count_diff: +1, rook_count_diff: –2,other material differences: 0,captured_score_diff: 0,capture_diff: 0,check_diff: +1, opening: Italian Game

The Elastic Net model estimates 0.631104 63.11% probability that white wins the game.

4. Interpretation of Results

In the begging of this project, I was interested in queen_count_diff and the elo_diff parameters therefore I would be interpreting some of the parameters that were selected by best model and one opening.

$$
\begin{aligned}
logit(\hat{p}) = 0.1878\ &-\ 0.1229\ \text{gameTypeClassical} +\ 0.0037\ \text{eloDiff} \\
&+\ 0.1973\ \text{queenCountDiff}\ +\ 0.0763\ \text{bishopCountDiff} \\
&+\ 0.261\ \text{capturedScoreDiff}\ +\ 0.0928\ \text{capturedDiff}\ +\ 0.0984\ \text{checkDiff} \\
&-\ 0.1101\ \text{openingItalianGame}\ \ldots
\end{aligned}
$$

Interpret the parameter estimate associated with queen_count_diff:

- 0.1973: As queen_count_diff increase by 1 piece, the expected log-odd of white winning the game increases by 0.1973, holding all other variable constant.

Interpret the parameter estimate associated with elo_diff:

- 0.0037: As elo increase by 1 point, the expected log-odd of white winning the game increases by 0.0037, holding all other variable constant.

Interpret the parameter estimate associated with game type Classical:

- -0.1239: The expected log-odds of white winning the game are 0.1239 lower for Classical than it is for Blitz game type, holding all other variable constant.

5. Evaluation and Discussion

The Elastic Net model was evaluated using cross-validation, which showed that a pure LASSO provided the lowest prediction error and the best overall performance rating difference, a subset of openings, and key material and tactical indicators such as queenCountDiff, bishopCountDiff, capturedScoreDiff, capturedDiff, and checkDiff improved the model ability to estimate the probability that white win.

But the model still has some limitations the dataset checks the state of the end of the game so there may be moves or some tactical variablabes for example the elastic model shrink the pawn count variables while having a pawn advtange it still valid indicture tword wining if we considered the promote of pawn this might give it more significant,

Other limitation is not beign able to include the draw games a limitation of used glm model that having the draw outcome may provide different kind of insigt.